ECE 143 · GROUP PROJECT

Quantitative Forecasting: Sportsbook Calibration & Prediction Market API Constraints

Evaluating NBA moneyline odds accuracy across 19,820 games (2008–2022)

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Why Do Sportsbook Probabilities Matter?

- Sportsbooks embed a "vig" (profit margin) into every line — odds sum to >100%
- Removing the vig reveals the sportsbook's true implied probability for each team
- If well-calibrated, these are the best publicly available pre-game forecasts

Raw Implied
Home −110 → 52.4%
Away −110 → 52.4%
Total = 104.8%

→

Overround (Vig)
104.8% − 100%
= 4.8% margin
✕ **Remove this**

→

Fair Probability
Home: 50.0%
Away: 50.0%
Sum = 100% ✓

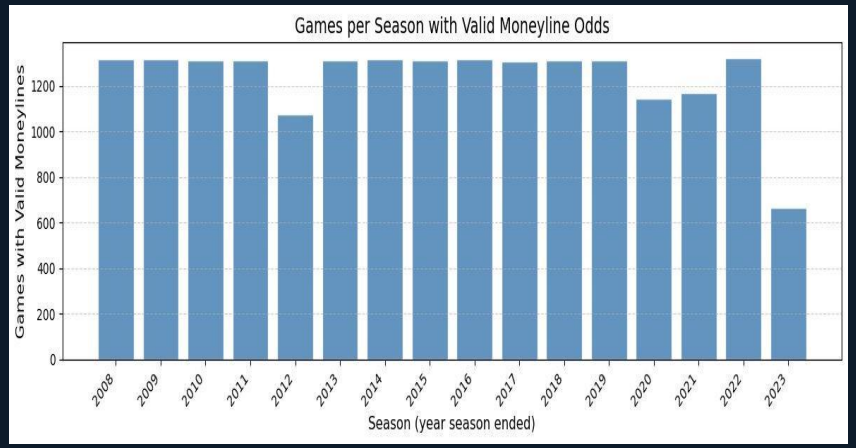
Research Question: *Are no-vig sportsbook probabilities genuinely well-calibrated against NBA game outcomes — and where does their accuracy break down?*

Dataset — Kaggle NBA Sportsbook Odds

23,118 total games (2007–2025)

19,820 valid moneyline odds (85.7%)

14 seasons fully covered (2008–2022)



Games with valid moneylines per season. 2023 is partial; 2024–2025 have none.

- Columns: moneyline, spread, total, quarter scores, game type
- Average sportsbook vig: 3.77% (range: −3.9% to 73.3%)
- 2024–2025 seasons have no moneylines in this dataset

2012 & 2020: shortened seasons
(lockout + COVID bubble)

Methodology — Building the Data Pipeline

Kaggle CSV (23k games)

Standardise names + event key

Polymarket Gamma API

Paginated fetch + local cache

Fuzzy Matching — rapidfuzz + 30-team alias (90% threshold)

Temporal Merge — pd.merge_asof (T−1 hr snapshot)

master_events.csv (512 matched games)

→ Then we hit a problem (next slide)

Pipeline complete — but Polymarket returned no usable historical data. We pivoted to sportsbook-only evaluation.

Engineering Finding — Polymarket API Constraints

What We Attempted

→ Fetch Polymarket NBA market prices at T−1 hour before each game's tip-off

→ Compare calibration: centralised sportsbook vs decentralised prediction market

→ A novel side-by-side comparison using proper scoring rules (Brier, Log Loss)

What We Found

Historical data purged

/prices-history returns {"history": []} for closed markets

Terminal prices only

outcomePrices are 0.0 or 1.0 for closed events — not pre-game

No individual game markets

Only season-long futures found — no per-game NBA markets

Conclusion: Open Prediction market API's dont tend to have historical data related to games, but we go with private API's we might find significant data.

Methodology — No-Vig Conversion & Scoring

No-Vig Conversion

Step 1: American odds ⟶ Implied probability

```
-110 → 110/(110+100) = 52.4%
```

Step 2: Home + Away implied = Overround

```
52.4% + 52.4% = 104.8% (the vig)
```

Step 3: Fair prob = Implied / Overround

```
52.4% / 104.8% = 50.0%
```

Step 4: Evaluate fair_prob_home vs outcome

```
1 = home win, 0 = away win
```

Scoring Rules

Brier Score

mean($(p - y)^2$)
Range: 0 (perfect) ⟶ 1 (worst)
Baseline: 0.25 (always 50/50)

Log Loss

Penalises confident wrong predictions harder than Brier

Reliability Diagram

Predicted prob (x) vs actual win rate (y). Perfect = diagonal

Baseline 0.25 — the number to beat

Finding 1 — Sportsbooks Are Well-Calibrated

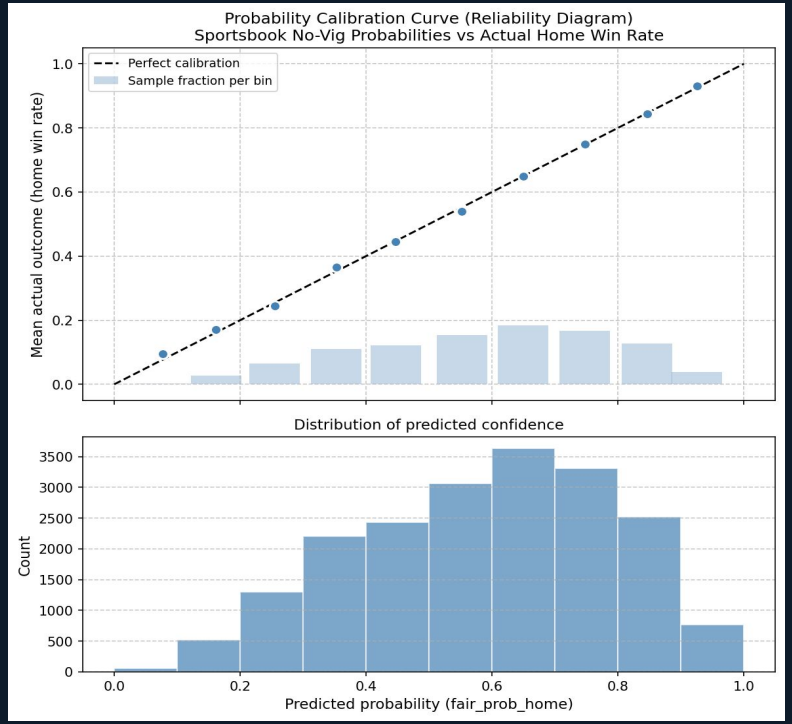
"When the sportsbook says 70% — do teams actually win 70% of the time?"

Answer: Yes, closely.

0.2024 — **Brier Score**
95% CI [0.2000, 0.2047]

0.2500 — **Random baseline**
always predict 50/50

~19% — **Better than random**
19,820 games evaluated



Points hug the diagonal — predicted probabilities closely match actual win rates across the full range.

Key: Dashed diagonal = perfect calibration. Blue dots = sportsbook predictions. The closer to the line, the more accurate.

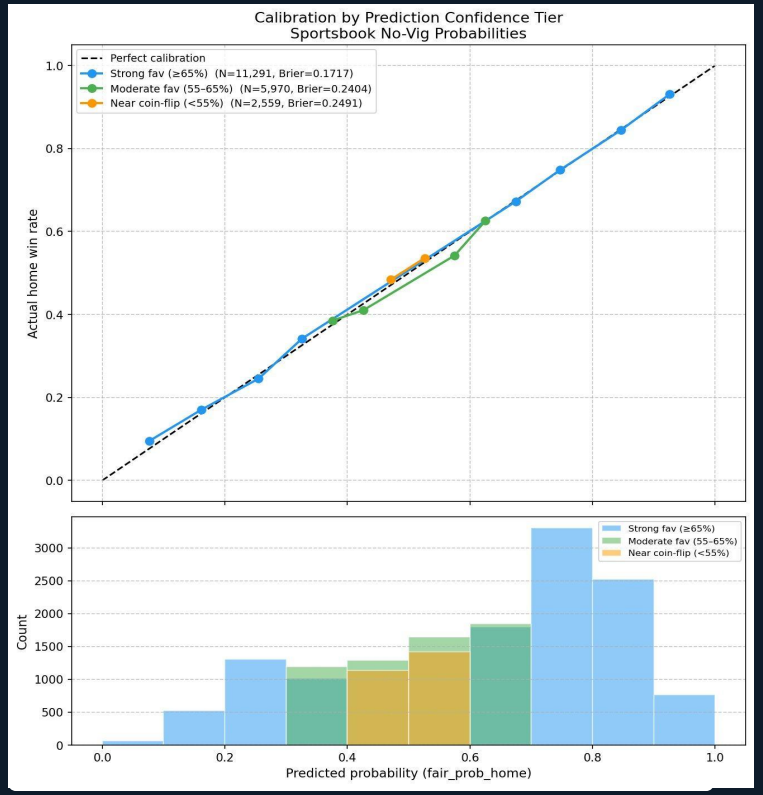
Finding 2 — Accuracy Depends on Confidence

"Is the sportsbook equally accurate for blowouts vs coin-flip games?"

Answer: No — accuracy drops sharply for close games.

Tier	Brier	N
Strong fav (≥65%)	**0.1717**	11,291
Moderate (55–65%)	0.2404	5,970
Near coin-flip (<55%)	**0.2491**	2,559
Random baseline	**0.2500**	—

Near coin-flip Brier (0.2491) ≈ random (0.2500)
→ No meaningful edge in close games



Blue = strong favourite. Orange = near coin-flip. Note how orange barely deviates from the diagonal.

Finding 3 — Playoffs Are Marginally Harder

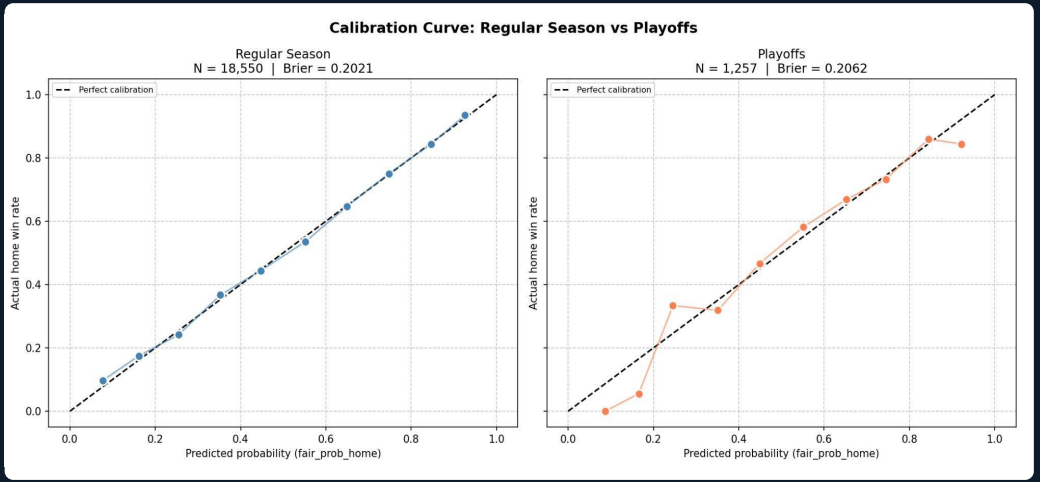
"Does calibration hold in the playoffs, where matchups are closer and strategies more complex?"

0.2021 — **Regular Season**
N = 18,550

0.2062 — **Playoffs**
N = 1,257

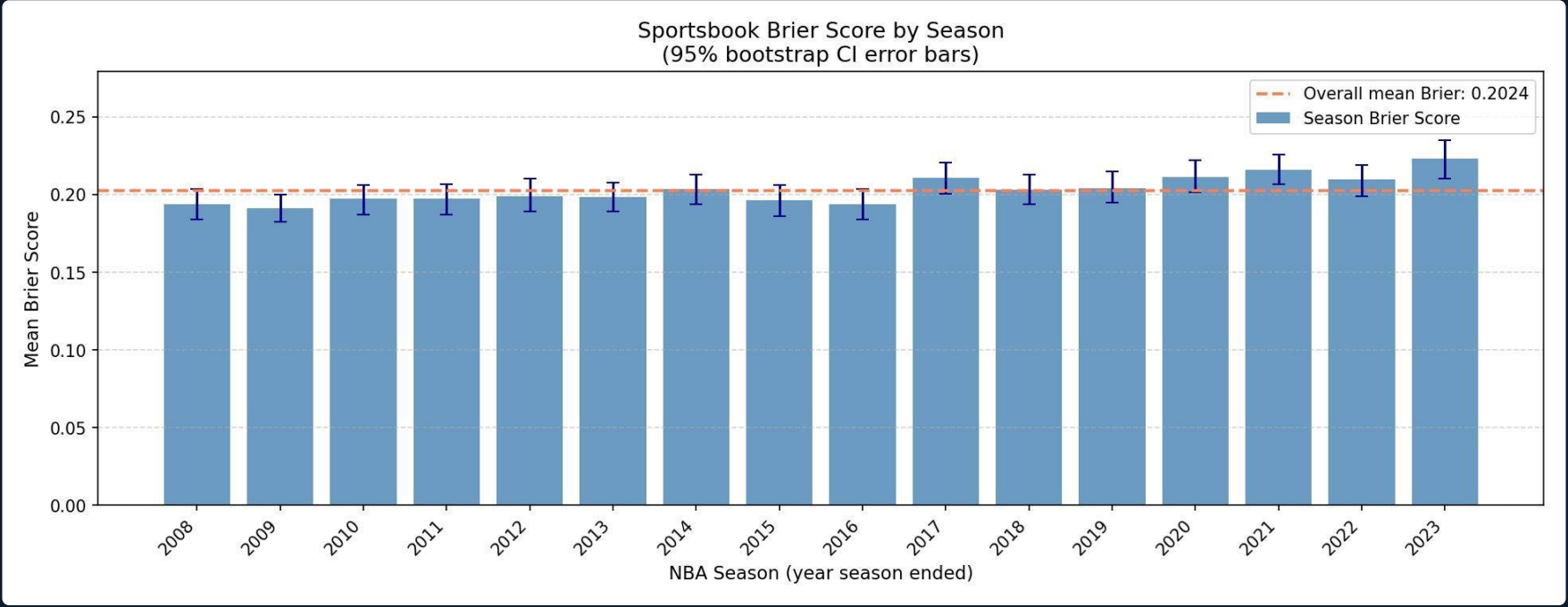
Difference: +0.0041 (small)

Playoff sample is 14× smaller — interpret cautiously.
CIs overlap — the difference is not statistically significant.



Left: Regular Season. Right: Playoffs. Both track the diagonal closely — calibration holds.

Finding 4 — Stable Calibration Since 2008



No systematic trend across 14 seasons
95% CI bars overlap the overall mean in most seasons
⟶ stable, mature, efficient market

2023 shows slight uptick (0.22)
But it's a partial season (fewer games) and CI is widest ⟶ noise, not signal

Conclusions

Sportsbooks are well-calibrated overall

Brier Score 0.2024 — ~19% better than random (0.25)
Stable and consistent across 19,820 games and 14 seasons

Predictive edge is concentrated in clear mismatches

Strong favourites (≥65%): Brier 0.1717 ← real signal
Near coin-flip games (<55%): Brier 0.2491 ← no edge

Polymarket API does not retain historical pre-game prices

Decentralised markets are not currently viable for
retrospective probability analysis — an open research gap